

AHMAD SYAFIQ KAMIL

Jakarta, Indonesia

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SUMMARY

Software engineer with 7+ years of experience building AI-powered systems and scalable data infrastructure for mission-critical applications. Proficient in cloud-native deployment with AWS, Kubernetes, and Python, with a track record of processing large-scale datasets in high-stakes environments including diplomatic missions and national research institutions. ISO 27001 certified with hands-on experience in data security and compliance. A generalist engineer who rapidly adopts new technologies in fast-paced environments, currently integrating AI-assisted development workflows to accelerate delivery. Motivated to apply technical expertise to impactful problems at scale, with a growing focus on healthcare technology and patient outcomes.

EXPERIENCE

Syamil Solusindo Semesta

Senior Full Stack Engineer & Lead Consultant

Remote / Jakarta

Jan 2022 – Present

Led end-to-end design, development, and deployment of 8 mission-critical applications across 7 Indonesian diplomatic missions, serving 500+ daily active users and 1M+ registered WNI.

- Designed and deployed a suite of 8 applications across 7 diplomatic missions maintaining 99.9% uptime, serving a registered base of over one million Indonesian citizens.
- Built consular service and queue management systems for KJRI Dubai using Next.js, Django REST Framework, and MySQL, streamlining citizen-facing service delivery.
- Building a production-ready AI chatbot for consular services using Google Agent Development Kit (ADK), FastAPI, and PostgreSQL — one of the first LLM-agent deployments in Indonesian diplomatic infrastructure.
- Implemented a multi-mission consular protection platform across 6 Indonesian missions in Malaysia using PHP, CodeIgniter, and MySQL, handling citizen registration, legal protection, and welfare tracking.
- Containerized all mission applications using Docker on dedicated on-premise servers across 7 missions, reducing maintenance downtime by 40% and achieving 99.9% uptime.
- Implemented audit logging, RBAC, and AES-256 encryption for all applications handling classified diplomatic and citizen data, in compliance with ISO 27001 standards.

Key Achievements

1. Protkons platform deployed across 6 Malaysian missions serves over one million registered WNI with consular protection and welfare services.
2. Achieved consistent 99.9% uptime across all 7 diplomatic mission deployments through Docker-based containerization and proactive infrastructure management.
3. Leading deployment of an LLM-agent chatbot for KJRI Dubai among the first AI-powered citizen services in Indonesian diplomatic infrastructure.

Badan Riset dan Inovasi Nasional (BRIN)

Perekayasa (Research Engineer)

Jakarta

2022 – 2025

Contributed to national policy research by building data tools, automation pipelines, and analytical systems to support evidence-based policymaking across multiple government ministries.

- Developed automated web scraping tools using Python (BeautifulSoup, Selenium) to collect 10,000+ rows of structured data, significantly reducing manual effort for research teams.
- Transformed raw datasets into actionable insights for policy memoranda submitted to multiple government ministries, using Python (Pandas, Matplotlib, Seaborn) and Excel.
- Built an automated speech-to-text transcription tool in Python that reduced manual transcription time for field recordings and qualitative research interviews.
- Developed a web app integrating the BPS Open API to allow researchers to programmatically search, filter, and retrieve national statistical data.
- Coordinated regularly with BRIN researchers and officials from multiple ministries to bridge technical implementation and policy research requirements.

Key Achievements

1. Data analysis outputs directly contributed to policy memoranda (naskah kebijakan) submitted to multiple Indonesian government ministries and agencies.
2. BPS API integration app adopted as internal research infrastructure, enabling programmatic access to national statistics for BRIN research teams.
3. Automated data collection tools reduced manual research data gathering effort across multiple ongoing national research projects.

Upwork (Freelance)

Full Stack Developer & DevOps Engineer

Remote
2023 – Present

- Deployed and configured ODK Central instances on DigitalOcean using Docker Compose for 2 international clients, enabling reliable mobile data collection for field research.
- Contributed to a medical data collection project for a hospital in Bandung, preparing ODK Central infrastructure as part of a health survey initiative led by PCNU Germany.
- Provided clear setup documentation and post-deployment support so clients could independently manage their ODK Central environments.

Badan Pengkajian dan Penerapan Teknologi (BPPT)

AI/ML Engineer

Jakarta
2021 – 2022

- Developed a production-grade face recognition system using Python, TensorFlow, and ResNet achieving 92% accuracy, deployed as an attendance application currently in active use by a government ministry.
- Built a native Android application integrated with the face recognition backend API, enabling real-time face-based attendance capture in the field.
- Co-built an end-to-end MLOps infrastructure from server setup to CI/CD pipelines using Kubernetes, enabling automated model training, versioning, and deployment.
- Conducted hands-on training for 20+ research engineers on operating the MLOps infrastructure, accelerating team adoption of containerized ML workflows.

Embassy of Indonesia in Kuala Lumpur (KBRI)

Software Engineer

Malaysia
2019 – 2021

- Built and deployed 3 applications (Numbering System, Queue Management System, COVID-19 Aid Distribution System) in ~14 months using Django and MySQL.
- Managed transmission of ~50 diplomatic messages/telegrams daily to and from KBRI, and provided IT support for embassy staff resolving system and network issues.
- Digitized manual workflows, reducing document processing time by 30% and improving service delivery for embassy staff and citizens.
- Maintained 10+ embassy IT systems and networks with 99% uptime while coordinating with external vendors and managing on-premise servers.

Universitas Nahdlatul Ulama

Lecturer, Computer Science

Surabaya
2018 – 2019

- Taught 24 credit hours weekly across core CS courses: Algorithms, Data Structures, Web Development, and Mobile Programming.
- Mentored 100+ students per semester with curriculum aligned to industry standards.

KEY PROJECTS

AI-Powered Consular Chatbot — KJRI Dubai

In Development, 2025–2026

Building a production-ready AI agent for consular services using Google ADK, Python, FastAPI, and PostgreSQL. One of the first LLM-agent deployments in Indonesian government diplomatic infrastructure, enabling 24/7 automated citizen services for WNI in UAE.

Tech: Google ADK, Python, FastAPI, PostgreSQL, Docker

Protkons — National-Scale Consular Protection Platform

2022–Present

Multi-mission consular protection system deployed across 6 Indonesian missions in Malaysia, serving 1M+ registered WNI. Handles citizen registration, legal protection, welfare tracking, and case escalation.

Tech: PHP, CodeIgniter, MySQL, Docker

Centralized Service Data Dashboard

2023

Unified data intelligence platform aggregating service metrics from 6 Indonesian missions in Malaysia into a single dashboard for leadership decision-making and data-driven policy.

Tech: Laravel, PostgreSQL, Docker, Nginx

MLOps Infrastructure for National Research

2022

Architected shared machine learning infrastructure using Kubernetes and Kubeflow, adopted by multiple research centers for deploying computer vision models at scale.

Tech: Kubernetes, Docker, Python, TensorFlow, MLflow

AI-Powered Data Pipeline Automation

2025

Built a working prototype of intelligent data processing workflows using Claude AI API and n8n automation platform. Explored AI-driven data validation and automated reporting for research datasets.

Tech: Python, Claude AI, n8n, Docker, PostgreSQL

BetterFuture & E-Voting DApps

2025

Developed decentralized applications using Solidity smart contracts and Vite.js frontend with secure on-chain voting mechanisms and tokenized incentive systems.

Tech: Solidity, Ethereum, Vite.js, Web3.js, Hardhat

Parvata & FOCO iOS Apps

2024

Native iOS productivity and focus management tools built with Swift and SwiftUI, featuring CoreData integration, background notifications, and Apple HIG-compliant interfaces.

Tech: Swift, SwiftUI, CoreData, CloudKit

EDUCATION

Master of Computer (M.Kom), Information Systems

2015 – 2018

Institut Teknologi Sepuluh Nopember (ITS) Surabaya | GPA: 3.38

Bachelor of Computer (S.Kom), Information Systems

2010 – 2015

Universitas Jember | GPA: 3.34

SKILLS

- **Cloud & Infrastructure:** Kubernetes (K8s), Docker, AWS, CI/CD Pipelines, DigitalOcean, Linux Administration, Nginx
- **Languages:** Python (Advanced), TypeScript/Javascript, Go, SQL, PHP, Rust, Solidity
- **Backend & Frameworks:** Django REST Framework, FastAPI, Flask, Laravel, CodeIgniter, Node.js, Next.js, React
- **Data Engineering:** Pandas, NumPy, dbt, dlt, Airflow, Data Warehousing, ODK Central, MySQL, PostgreSQL
- **AI & Machine Learning:** Claude, Google Agent Development Kit (ADK), GitHub Copilot, LangChain, MLOps, Computer Vision (OpenCV), n8n Automation
- **Security & Compliance:** ISO 27001 Certified, RBAC Implementation, AES-256 Encryption, HIPAA-aligned data handling
- **Tools & Collaboration:** Claude Code, Cursor, Git, GitHub Actions, Postman, Figma, Jira, Slack, Agile/Scrum

CERTIFICATIONS & TRAINING

- KodingDeepDive x BwAI Jakarta - Architect AI Agent Systems with Agent-Database Lifecycle (2025)
- Langchain Academy: Deep Agent with Langgraph (2025)
- SUI Blockchain Bootcamp (2025)
- Blockchain Developer (Solidity) — BlockDevId (2025)

- Apple Academy Developer (2024)
- DigitalOcean Kubernetes Challenge (2023)
- Digitalent Big Data — Ministry of Communication and Information Technology (2022)
- Lead Auditor ISO 27001:2013 — Information Security Management (2019)

ADDITIONAL INFORMATION

- Work Authorization: Indonesian citizen, eligible to work remotely for US companies without visa sponsorship.
- Timezone Flexibility: UTC+7, experienced with remote collaboration across timezones (Malaysia, UAE, US-based clients). Comfortable adapting for US East Coast overlap.
- Languages: English B2 Upper Intermediate (Professional Working Proficiency), Bahasa Indonesia (Native).
- Availability: Open to occasional travel to NYC headquarters for team collaboration and onboarding.